

VELMÈRE BASIC AUDIT REPORT

Arbitrum Delayed Inbox (0x4dbd4fc535bd2916850904481a1be141421bd113)

VELMERE SECURITY AUDIT REPORT: ARBITRUM DELAYED INBOX

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: undefined | Tier: BASIC | Application Surface: CANONICAL

Created At: 2026-09-09T19:29:42.056Z

VERDICT SUMMARY:

ELEVATED RISK (UNVERIFIED CODE) (72/100)

Release Decision: BLOCKED | Verification Status: INSUFFICIENT_EVIDENCE

STOP-SELL DIRECTIVE: INSUFFICIENT_EVIDENCE: Coverage below minimum threshold for institutional sale

Confidence Score: 60/100 Evidence Coverage: 20%

Merkle Root: sha256:30f576046f490d462592455eff9bb1faafa90097b7bcb4d5ecc18f8b93fb18f9

Contract lacks verified on-chain runtime bytecode. In accordance with Velmère evidence standards, missing bytecode results in an elevated risk assessment (72/100).

AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Arbitrum Delayed Inbox
Contract Address: 0x4dbd4fc535bd2916850904481a1be141421bd113
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: ARB-INBOX
Proxy Pattern: unverifiable
Compiler Version: Unverified Source

SOURCE CODE VERIFICATION & BYTECODE DECOMPILEATION

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode unavailable. Disassembly, selector analysis, and privileged opcode scans cannot proceed.

BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

Findings & Vulnerability Analysis:

HIGH

VLM-EVM-NO-CODE: Unverified On-Chain Code / Missing Bytecode

Category: Missing Bytecode Truth | State: recommended
Evidence: Deployed runtime code is empty or unverified on target chain.
Recommendation: Deploy and publish verified source code or submit exact deployed bytecode before interaction.

PERMISSION & ROLE GOVERNANCE MAP (PRO)

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

PRO

Summary of Analytical Depth in this section:

- Admin role: privilege and access control hierarchy in bytecode
- Verification of timelock delay parameters and governance controls
- Detection of privileged mint, pause, blacklist, and drainage capabilities
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

LIQUIDITY HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

PRO

Summary of Analytical Depth in this section:

- Token distribution curve and top holder concentration analysis
- DEX liquidity pool pairing and lock verification across registered vaults
- Automated honeypot simulation, transfer fee checks, and sell restriction audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.

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ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)		PRO
[LOCKED]	[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]	

Summary of Analytical Depth in this section:

- Cross-function and cross-contract reentrancy guard modeling
- Price oracle staleness, manipulation resilience, and market feed verification
- Economic shock modeling and flash-loan manipulation resistance simulation
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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STORAGE LAYOUT COLLISION & MULTICOMPILER BYTECODE DIFF (ADVANCED)		ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]	

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)		ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]	

Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): PROPERTY P VERIFIED UNDER ASSUMPTIONS A FOR MODEL M
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)		ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]	

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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